# 课程作业1

C语言游戏

（飞机大战/贪吃蛇）

# 基本规则

- 两个游戏选一个实现
- 交作业形式：
  - 提交书面报告和代码，报告包括设计理念、界面展示等。

  [更多信息涉及课程隐私内容]
- 不可抄袭

# 空战游戏

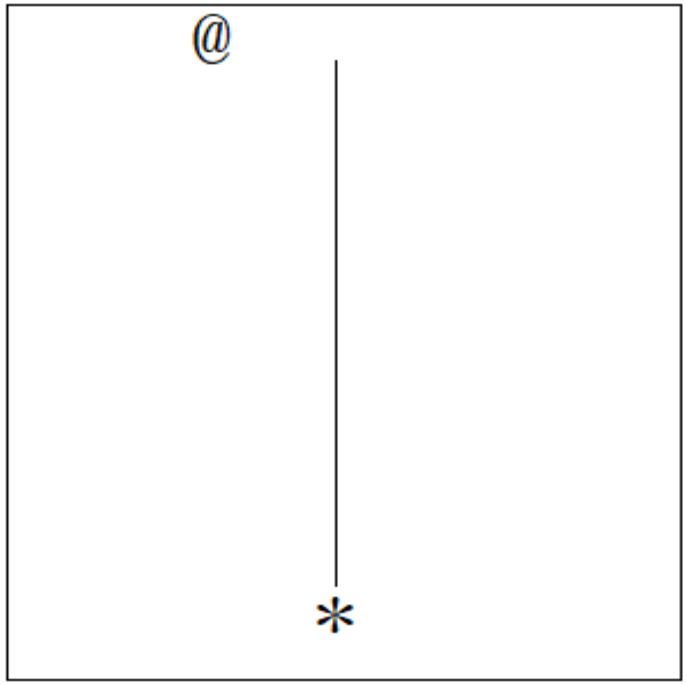
-- 数组、函数、三种控制语句的综合应用

# 目标

- 实现空战游戏
  - 实现基本的打敌机功能
  - 实现多台移动敌机、计分功能
  - 至少实现一个自己设计的游戏规则
- 鼓励实现独特和漂亮的外观效果

# 飞机的显示与控制

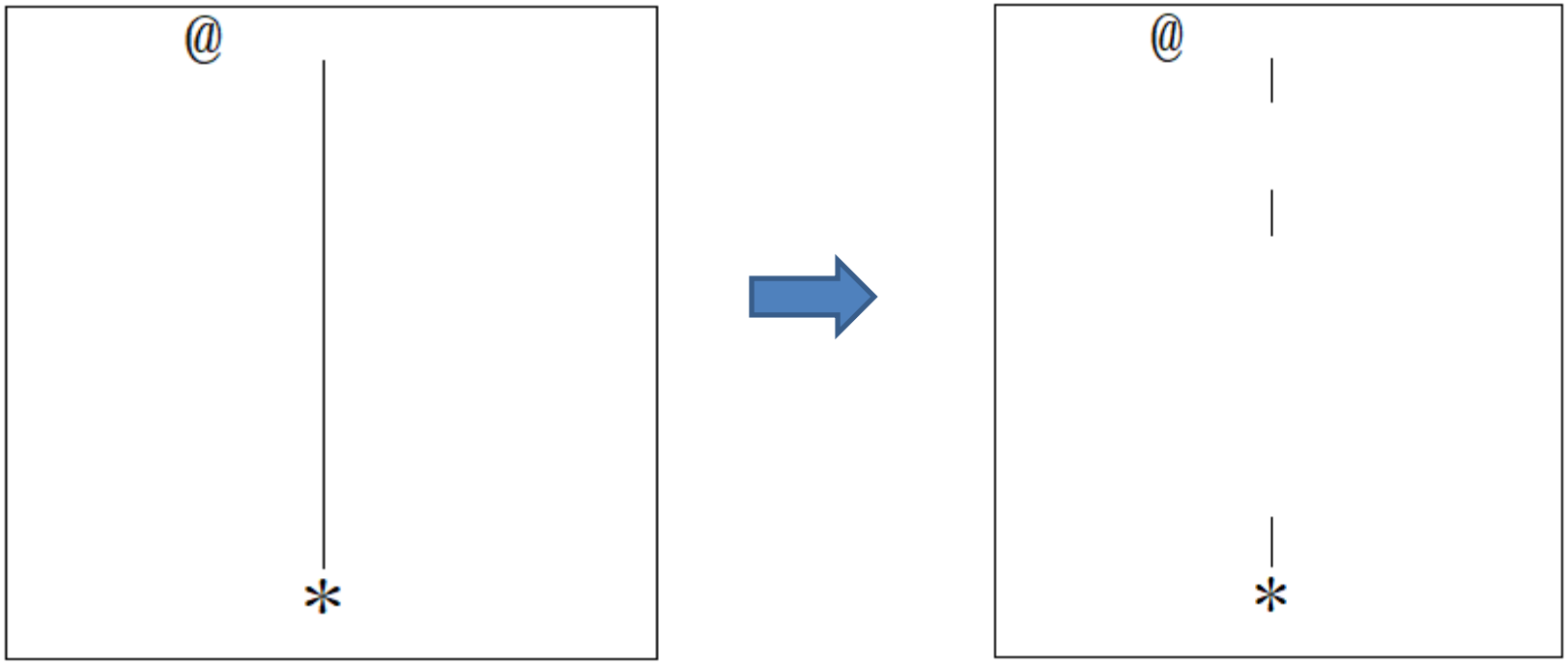
- 在二维数组 int canvas[High][Width]中存储游戏画面数据




| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 3 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 2 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 2 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 2 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 2 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 2 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 2 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |

# 子弹的形态

- 一竖线/逐渐上移的短线
- 可连续按空格键，显示多发子弹



# 敌机下移

- 敌机以固定速度向下移动
  - 如何控制敌机的位置？
  - 如何控制下降速度？

# 计分并输出

- 增加变量score来记录分数
  - 如果子弹击中一架敌机，加一分
  - 如果一架敌机移动到游戏画布范围以外，减一分，重新随机生成一架新的敌机
  - 如果敌机撞到我机，游戏结束退出

# 多架敌机

- 使用数组来存储多加敌机的位置

# 贪吃蛇

-- 数组、函数、三种控制语句的综合应用

# 目标

- 实现传统的贪吃蛇游戏
  - 移动、吃食、变长
- 至少实现一个自己设计的游戏规则
- 鼓励实现独特和漂亮的外观效果

```
##############################
#                            #
#                            #
#                            #
#                            #
#                            #
#                            #
#                            #
#                            #
#                            #
#                            #
#              ****@         #
#                            #
#                            #
#                  F         #
#                            #
#                            #
#                            #
#                            #
#                            #
##############################
```

# 构造小蛇

- 在二维数组 int canvas[High][Width]中存储游戏画面数据

```
##################################
#                                #
#                                #
#                                #
```

# 小蛇的移动（关键）

- 移动前后数组状态的变化
- 例子1. 向右运动

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 4 | 3 | 2 | 1 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | -2 | 0 | -1 |
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 4 | 3 | 2 | 1 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | -2 | 0 | -1 |
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |

# 小蛇的移动（关键）

- 移动前后数组状态的变化
- 例子2. 向上运动

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 4 | 3 | 2 | 1 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | -2 | 0 | -1 |
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 4 | 3 | 2 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | -2 | 0 | -1 |
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |

# 小蛇的移动（关键）

- 数组中小蛇对应元素的变化规律：
  1. 所有元素加1
  2. 最大值的元素置零（去掉原蛇尾）
  3. 根据移动方向，设置新的蛇头位置为1

- 向上移动一次所对应的三步操作举例

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 4 | 3 | 2 | 1 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | -2 | 0 | -1 |
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |

加一

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 5 | 4 | 3 | 2 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | -2 | 0 | -1 |
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |

去尾

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 4 | 3 | 2 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | -2 | 0 | -1 |
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |

加头

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 4 | 3 | 2 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | -2 | 0 | -1 |
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |

# 小蛇的移动（关键）

- 实现：
  - 定义 moveSnake () 函数，其中：
    - 定义变量来区分上下左右四种移动
    - 使用条件语句来分别实现四个方向的移动
  - 在updateWithInput()中调用上述函数，实现用户控制
  - 在updateWithoutInput()中调用上述函数，实现用户未输入时小蛇的移动

# 小蛇的移动（关键）

- 特殊情况：
  - 小蛇撞自己的身体会怎么样？
  - 小蛇撞到画布边缘会怎么样？
  - 还有什么特殊情况？

# 吃食物增加长度

- 思路和移动类似，只需保留蛇尾，不将最大值变为0即可

# 吃食物增加长度

- 例子：

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | -2 | 0 | -1 |
| -1 | 0 | 0 | 0 | 4 | 3 | 2 | 1 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |

加一

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 5 | 4 | 3 | 2 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |

加头

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | -1 |
| -1 | 0 | 0 | 0 | 5 | 4 | 3 | 2 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -1 |
| -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |